Operational Research

5CM522



5CM522 Assessment Brief – Coursework

Part A

Dr Sam O’Neill

Sensitivity: Internal

Operational Research (5CM522)

Contents

Module Leader 3
Key dates and details 3
Description of the assessment 3
Learning Outcomes 3
Assessment Regulations 3
Guidance and Presentation 4
Submission Requirements 4
Viva Voce (Oral Examination) 4
Questions 5
Appendix A 8
    Presentation Rubric 8

Module Leader

- Dr Sam O'Neill
- Contact via the module page

Key dates and details

Assessment Type:	Individual
Assessment weighting:	25%
Learning Outcomes:	1, 2
Submission Method:	Electronic Submission via Course Resources
Submission Date:	12/12/25
Provisional Feedback Release Date:	16/01/26

Description of the assessment

Students will be required to model and answer a series of questions related to Operational Research

Learning Outcomes

1. Demonstrate an understanding of Operational Research methods and methodology.
2. Build models to enable the analysis and solution of case study problems.

Assessment Regulations

The University's regulations, policies and procedures for students define the framework within which teaching, and assessment are conducted. Please make sure you are familiar with these regulations, policies, and procedures.

Guidance and Presentation

Marks are given for presentation based on the following weighting:

- 15% for presentation and quality of write up. Please see the rubric in Appendix A.
- 85% for answers given in the report.

You are expected to write up your models and answers. This includes defining notation, e.g. variables, constants etc.

You should try and use the built-in equation editor in word, or Latex; however, you may provide handwritten answers that are scanned into one document. Please ensure that any handwritten answers are neat and legible.

Submission Requirements

Students are required to submit a word/pdf document that contains all their answers to the questions. They should also submit any Python code used for the solutions to the models as separate files that are clearly labelled with the question no. If these are .ipynb (Google Colab) files then you can download the notebook and submit this.

Screenshots are encouraged to show the output of your Python code and solutions to the problems.

These should all be submitted to the submission point before the deadline.

Viva Voce (Oral Examination)

Students will be asked to demonstrate their work over a 10-15 minute period. This is to ensure that the work submitted is the student's own and inline with the academic assessment regulations and policy on Generative AI.

Assessment Rubric

Please see the assessment rubric section.

Assessment AI-assistance

You are not permitted to use Generative AI for this assignment.

Any suspicion that the work is not your own, either through plagiarism, excessive collaboration or use of generative AI will result in a further Oral Examination. This will require you to explain your work in detail, failure to do so will result in a Fail mark.

Any misuse of generative AI tools could be considered ethical academic misconduct (as per Academic Regulations, Academic Misconduct, Section J2). If in doubt, please consult your module leader.

Questions

1. For the following linear program:

maximise

subject to:

a) Graph the linear programming problem and solve by inspecting the basic feasible solutions via each possible pair of simultaneous equations formed by the constraints. Show your working.
b) Which constraints are binding?
c) Convert into the following matrix form.

minimise

subject to:

d) Solve the linear program using Python.

(30 marks)

2. You have a total of £10,000 to invest in three different funds - ToyTown, C&G and Pixel Ltd. The expected return and risk level for each of the fund is given in table 1:

Fund	**Expected Return**	**Risk Level**
Toy Town	6%	Low
C&G	7%	Medium
Pixel Ltd	10%	High

Table 1:Expected return and risk level

To minimise your risk across the three funds you decide to invest at least twice as much in Toy Town as you invest in C&G. You also decide that as Pixel Ltd is considered a high-risk fund, to limit the amount you invest in this fund to £2000.

a) Taking account of your decisions to minimise the risk, produce a model to determine the optimal amount to invest in each of the three funds so as to maximise the expected return?
b) Solve this model using Python and report the solutions.
c) Reduce this model to only two decision variables. What does this tell you about the solution to the problem?

(30 marks)

3. The following items (see Table 2) are being given away for free and you wish to maximise the value of the items that you take away. Your rucksack has a maximum weight of 15kg and a maximum volume of 30 litres.

Item	Weight (Kg)	Volume (Litre)	Value (£)
1	3.5	4	5
2	2.4	4	2
3	0.9	5	3
4	2.5	2	1
5	3.1	11	11
6	4.2	9	7
7	5.1	11	5
8	3.5	8	2
9	4.5	11	5
10	2.9	8	7

Table 2: Item information

a) Formulate an Integer Linear Program (ILP) with appropriate notation to solve the problem.
b) Solve this ILP using Python.
c) Relax the integer constraints of the ILP and solve the LP.
d) Round your solution from c) to the nearest integer and compare it with the integer solution from b). How does this compare and what can you say integer programming and linear programming problems?

(40 marks)

Appendix A

Presentation Rubric

Presentation Aspect	Excellent (5)	Very Good (4)	Good (3)	Fair (2)	Poor (1)	Inadequate (0)
Organisation and Structure	The report is exceptionally well-organised, with a clear and logical structure that guides the reader through the optimisation process. Each section is clearly labeled, and there is a logical flow from problem statement to solution.	The report is well-organised, with a clear structure that effectively guides the reader through the optimisation process. Sections are appropriately labeled and connected.	The report is organised and follows a logical structure, but there may be some minor issues with flow or clarity in the organisation.	The report is somewhat disorganised, making it challenging for the reader to follow the optimisation process. The structure may lack clarity.	The report lacks organisation and structure, making it difficult for the reader to understand the optimisation process.	
Clarity and Readability	The report is exceptionally clear and easy to read. It uses professional language, and sentences are concise and well-constructed. The report is free from grammatical and typographical errors.	The report is clear and highly readable. It uses professional language, and sentences are well-constructed. There may be a few minor grammatical or typographical errors.	The report is mostly clear and readable, but there may be some issues with sentence structure or word choice. A few grammatical or typographical errors may be present.	The report is somewhat clear but may require effort from the reader to understand the content. Sentence structure and word choice may hinder clarity. Several grammatical or typographical errors are present.	The report is unclear and challenging to read. Sentence structure and word choice significantly impede clarity. Numerous grammatical and typographical errors are present.	
Visual Aids and Equations	Visual aids, such as graphs, charts, and tables, are exceptionally well-incorporated to enhance understanding. Equations are presented using the equation editor or LaTeX, and they are neat, well-labeled, and easy to follow.	Visual aids are effectively used to enhance understanding, and equations are presented clearly using the equation editor or LaTeX. There may be a few minor issues with labeling or presentation.	Visual aids are used, but there may be some minor issues with their effectiveness or clarity. Equations are presented using the equation editor or LaTeX but may lack some neatness or labeling.	Visual aids are somewhat used, but they do not significantly enhance understanding. Equations may be presented without using the equation editor or LaTeX and may lack neatness or labeling.	Visual aids are not effectively used, hindering understanding. Equations are presented without using the equation editor or LaTeX and lack neatness and labeling.	